Portfolio of

NOURHAN QAWI

DATA ANALYST

Turning data into actionable insights

PROFESSIONAL SUMMARY

NOURHAN QAWI

DATA ANALYST

I started with curiosity and found my answers in data.
Today, I use data to uncover insights, tell meaningful stories, and support smarter decisions. My experience across banking, government, and consulting helped me see data not just as numbers, but as a way to understand problems and create real impact.
I enjoy connecting the dots, simplifying complexity, and turning insights into action. For me, data is more than analysis, it’s a tool to think better, decide better, and build better.

CORE COMPETENCIES

TECHNICAL & ANALYTICAL SKILLS

DATA VISUALIZATION: *TABLEAU, POWER BI, EXCEL*

PROGRAMMING*: PYTHON, SQL*

STATISTICAL ANALYSIS

CRITICAL THINKING & PROBLEM SOLVING

STRATEGIC & ANALYTICAL MINDSET

COMMUNICATION & STORYTELLING

TEAM LEADERSHIP & COLLABORATION

CREATIVITY & INNOVATION

ACTIVE LEARNING & ADAPTABILITY

IMPROVING PROFITABILITY & DELIVERY PERFORMANCE

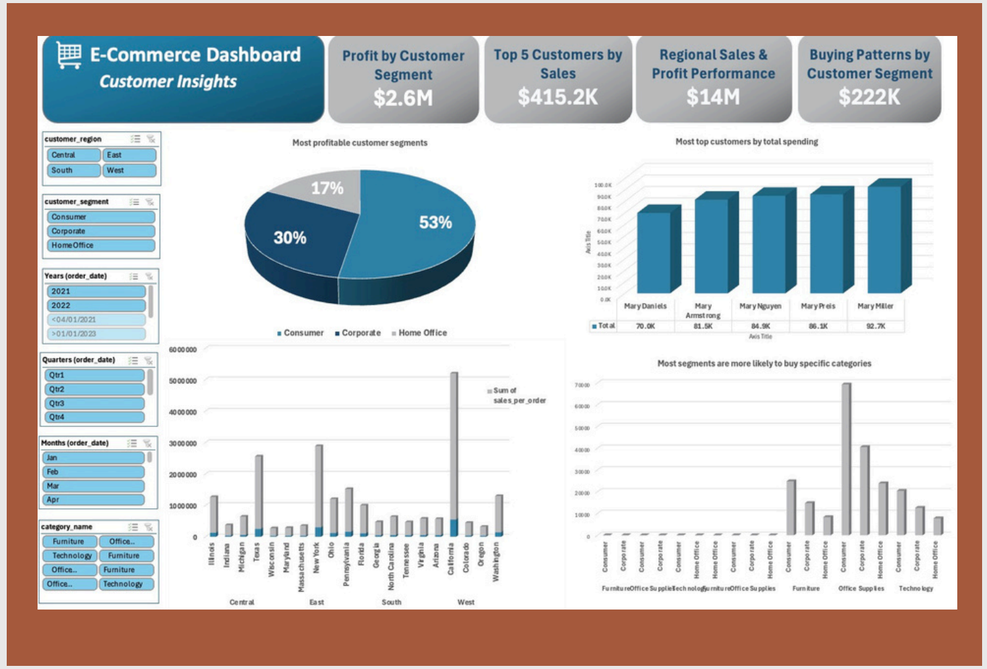
E-commerce Business Analysis

CASE OVERVIEW AND STRATEGIC OBJECTIVE

This case study examines an e-commerce business facing **inconsistent profitability despite maintaining strong sales volumes.** The strategic objective was to **explain why revenue growth was not translating into stable profits and to identify operational and behavioral drivers behind profit leakage**.

The analysis was framed around practical questions for operations managers, finance teams, and senior leadership. The business operates across multiple regions, product categories, and shipping methods, creating complexity in delivery performance and cost control.

Management observed rising customer complaints tied to late and canceled deliveries, alongside unexplained negative-profit orders. The goal is to **connect operational performance, customer behavior, and financial outcomes to support clear, actionable decisions centered on sustainability and long-term value.**



The presentation follows a narrative flow from context and problem definition to analysis, insights, and recommendations so stakeholders can understand not only what happened, but why it happened and what should be done next.

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BUSINESS CONTEXT & OPERATING ENVIRONMENT

Transactional e-commerce at scale

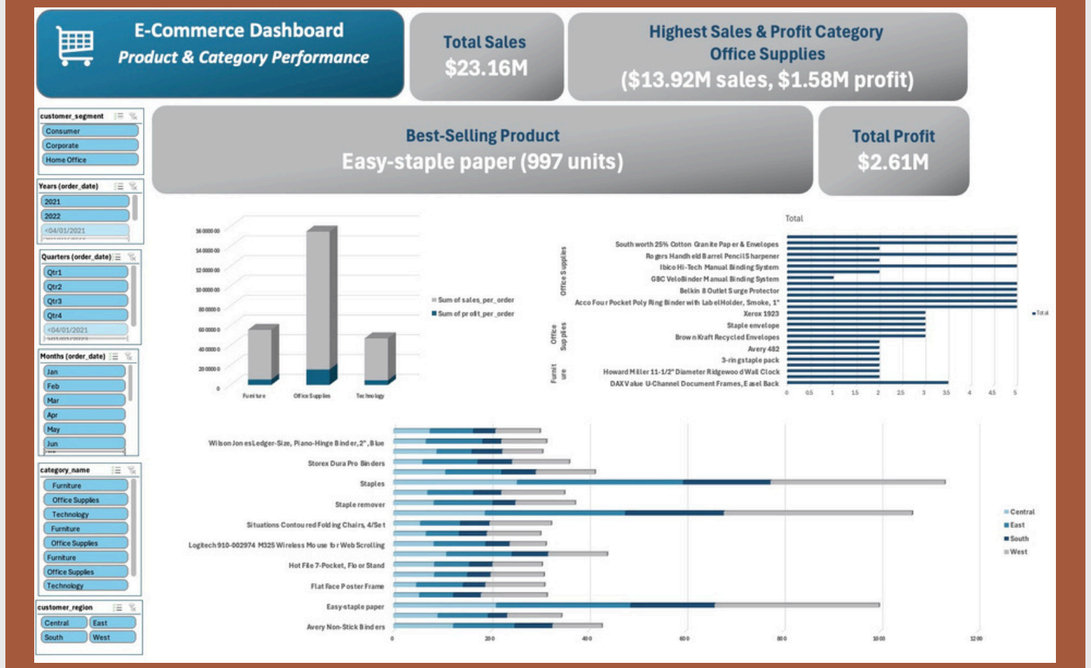
High order volume across Office Supplies, Furniture, and Technology; success depends on both sales and margin.

Fulfillment complexity

Multiple shipping classes and regional dispersion introduce variability in cost, reliability, and delivery timelines.

Profit volatility signal

Some “healthy” sales still generated losses, motivating deeper visibility into how delivery performance, customer type, and product mix interact.



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PROBLEM DEFINITION

Strong sales volumes, inconsistent profit outcomes

The core issue was a **disconnect between sales performance and profitability.** Overall sales volumes looked strong, yet profit results were uneven, with a meaningful share of orders producing negative profit.
At the same time, customer complaints about delivery delays and cancellations were rising, pointing to potential operational friction. Management lacked clarity on whether losses were isolated incidents or systemic patterns.
The analytical challenge was to pinpoint where and why profit was being lost, and to test whether operational factors such as delivery status and shipping type were materially affecting financial outcomes, moving from reporting to diagnosis for actionable decisions.

Total Profit
$2.61M

Total Sales
$23.16M

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KEY BUSINESS QUESTIONS GUIDING THE ANALYSIS

To address the identified challenges, the analysis was guided by a set of focused business questions.

These included: **where is profit being lost despite strong sales? How does delivery performance, including late and canceled orders, affect profitability? Are repeat customers more valuable and stable than new customers from a profit perspective? Which product categories and geographic regions contribute most to profit volatility?**

These questions reflect how analytics teams operate in industry, starting with hypotheses and information needs rather than tools or charts. By grounding the analysis in these questions, the work remained aligned with business priorities and ensured that findings could be translated into decisions rather than remaining abstract observations.

ANALYTICAL APPROACH AND STRATEGIC FRAMING

The analytical approach adopted in this case study was diagnostic and business driven. The analysis was structured around understanding relationships between operational performance and financial outcomes.
The goal was not only to describe what happened, but to uncover root causes and identify levers that management could realistically act upon.

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DATA SCOPE AND TOOL SELECTION

Cleaned, transaction-level e-commerce data (one row per order) with customer IDs, categories, delivery status, shipping type, dates, sales, and profit per order.

Excel was chosen to quickly aggregate, segment, and compare transactional patterns using pivot tables and calculated metrics.

customer_id	Customer Full Name	category_name	product_name	customer_segment	customer_city	customer_state	customer_region	delivery_status	order_date	order_id	ship_date	shipping type	days_for_shipment_scheduled	days_for_shipment_real	order_item_discount	sales_per_order	order_quantity	profit_per_order
C_ID_45866	Mary Fuller	Office Supplies	Xerox 1913	Corporate	New Rochelle	New York	East	Shipping on time	11/05/2022	O_ID_3001072	11/07/2022	Second Class	2	2	35	500	5	223.1999969
C_ID_44932	Alan Edelman	Office Supplies	#6 3/4 Gummed Flap White Envelopes	Corporate	HOUSTON	Texas	Central	Shipping canceled	20/06/2022	O_ID_3009170	23/06/2022	Second Class	2	3	85	500	5	199.1999969
C_ID_70880	Mary Gayman	Office Supplies	Belkin 8 Outlet Surge Protector	Consumer	Louisville	Kentucky	South	Late delivery	25/06/2022	O_ID_3047567	30/06/2022	Standard Class	4	5	75	44	5	195.5
C_ID_33157	Raymond Eason	Office Supplies	GBC VeloBinder Manual Binding System	Corporate	Chicago	Illinois	Central	Late delivery	10/06/2022	O_ID_3060575	10/10/2022	Second Class	2	4	60	254	1	220
C_ID_58303	Mary Gonzalez	Furniture	Eldon Pizzaz Desk Accessories	Home Office	Philadelphia	Pennsylvania	East	Late delivery	02/05/2022	O_ID_3064311	08/01/2022	First Class	1	2	125	500	1	97.5
C_ID_34556	Joe Leatherbury	Office Supplies	Hoover Shoulder Vac Commercial Portable Vacuum	Home Office	Columbia	South Carolina	South	Shipping on time	24/06/2022	O_ID_3074984	28/06/2022	Standard Class	4	4	80	500	2	-349.8599854
C_ID_56882	Mary Etezadi	Office Supplies	Rogers Handheld Barrel Pencil Sharpener	Home Office	Henderson	Kentucky	South	Advance shipping	07/10/2022	O_ID_3000007	16/04/2022	Standard Class	4	3	45	299.980011	5	71.40000153
C_ID_51638	Andrea O'Briant	Office Supplies	Dixon Prang Watercolor Pencils, 10-Color Set with Brush	Corporate	ATHENS	Georgia	South	Shipping on time	26/01/2022	O_ID_3000008	30/01/2022	Standard Class	4	4	45	299.980011	5	-31.87000084
C_ID_60578	Mary Etezadi	Office Supplies	Ibico Hi-Tech Manual Binding System	Home Office	Henderson	Kentucky	South	Shipping on time	09/03/2022	O_ID_3000009	09/07/2022	Standard Class	4	4	48	299.980011	5	118.4300003
C_ID_55200	Mary Etezadi	Office Supplies	Alliance Super-Size Bands, Assorted Sizes	Home Office	Henderson	Kentucky	South	Late delivery	19/03/2022	O_ID_3000010	25/03/2022	Standard Class	4	6	48	299.980011	5	78.87000275
C_ID_61171	Mary Etezadi	Office Supplies	Southworth 25% Cotton Granite Paper & Envelopes	Home Office	Henderson	Kentucky	South	Advance shipping	09/11/2022	O_ID_3000011	16/06/2022	Standard Class	4	2	48	299.980011	5	123.4700012
C_ID_50457	Richard Pak	Technology	Logitech K350 2.4Ghz Wireless Keyboard	Consumer	Madison	Wisconsin	Central	Late delivery	19/06/2022	O_ID_3079653	22/06/2022	Second Class	2	3	90	42	2	-88.97000122
C_ID_36240	Sarah Wiener	Furniture	DAX Value U-Channel Document Frames, Easel Back	Consumer	Seattle	Washington	West	Late delivery	11/05/2022	O_ID_3086366	11/11/2022	Second Class	2	6	27.5	196	5	217.3500061
C_ID_52640	Eugene Pryor	Technology	PowerGen Dual USB Car Charger	Home Office	Yonkers	New York	East	Late delivery	30/04/2022	O_ID_3000153	08/02/2022	Standard Class	4	5	0	499.9500122	2	149.9900055
C_ID_52478	Mary Ludtke	Office Supplies	Sensible Storage WireTech Storage Systems	Consumer	YONKERS	New York	East	Advance shipping	30/05/2022	O_ID_3000155	01/06/2022	Standard Class	4	2	20	499.9500122	2	36
C_ID_52183	David Sundaresam	Office Supplies	Acco Four Pocket Poly Ring Binder with Label Holder, Smoke, 1"	Consumer	Huntsville	Texas	Central	Shipping on time	03/02/2022	O_ID_3000016	05/04/2022	Standard Class	4	4	51	299.980011	5	-24.89999962
C_ID_28744	Mary Seite	Office Supplies	Newell 312	Consumer	Laredo	Texas	Central	Late delivery	27/02/2021	O_ID_3000017	04/03/2021	Second Class	2	5	4.800000191	119.9800034	2	-30.75
C_ID_45461	Mary Seite	Technology	Memorex Micro Travel Drive 8 GB	Consumer	Laredo	Texas	Central	Shipping on time	28/10/2021	O_ID_3000018	30/10/2021	Second Class	2	2	0.800000012	79.98000336	2	-122.7300034
C_ID_31115	Elizabeth Jacobs	Furniture	Howard Miller 11-1/2" Diameter Ridgewood Wall Clock	Corporate	Springfield	Virginia	South	Late delivery	02/04/2021	O_ID_3000019	07/05/2021	Second Class	2	6	4	100	2	33.59999847
C_ID_45766	Katherine Jacobs	Office Supplies	Avery 482	Corporate	Springfield	Virginia	South	Shipping on time	02/11/2021	O_ID_3000020	14/01/2021	Second Class	2	2	4	79.98000336	2	24.69000053
C_ID_47752	Mary Vernon	Furniture	DAX Value U-Channel Document Frames, Easel Back	Consumer	Dover	Delaware	East	Late delivery	01/12/2021	O_ID_3000021	15/01/2021	Second Class	2	3	9	100	2	9.100000381
C_ID_50054	Dennis Dahlen	Furniture	O'Sullivan Elevations Bookcase, Cherry Finish	Consumer	San Francisco	California	West	Late delivery	03/01/2022	O_ID_3000022	04/05/2022	Second Class	2	5	13	100	2	-21.75
C_ID_36146	Mary DeCherney	Furniture	Global Highback Leather Tilter in Burgundy	Consumer	Mount Pleasant	South Carolina	South	Shipping on time	09/11/2021	O_ID_3000023	16/06/2021	Second Class	2	2	15.30000019	89.98000336	2	22.40999985
C_ID_41234	Frances Dahlen	Office Supplies	4009 Highlighters by Sanford	Consumer	San Francisco	California	West	Late delivery	27/08/2021	O_ID_3000024	01/09/2021	Second Class	2	5	19.79000092	109.9400024	2	25.23999977
C_ID_32090	Alice Moore	Office Supplies	Avery Metallic Poly Binders	Consumer	Newark	Ohio	East	Late delivery	16/04/2021	O_ID_3000025	22/04/2021	Second Class	2	6	2.400000095	119.9800034	2	30.5699997
C_ID_34773	Kyle Selesnick	Office Supplies	3-ring staple pack	Corporate	Bossier City	Louisiana	South	Late delivery	25/05/2021	O_ID_3000026	29/05/2021	Second Class	2	4	24	119.9800034	2	46.0699997
C_ID_44046	Brenda Selesnick	Office Supplies	Brown Kraft Recycled Envelopes	Corporate	Bossier City	Louisiana	South	Shipping on time	13/04/2021	O_ID_3000027	13/06/2021	Second Class	2	2	36	299.9700012	3	-184.7799988
C_ID_46414	Frank Selesnick	Office Supplies	Staple envelope	Corporate	Bossier City	Louisiana	South	Late delivery	14/08/2021	O_ID_3000028	16/11/2021	Second Class	2	5	6	119.9700012	3	38.75
C_ID_46599	Johnny Moore	Office Supplies	Xerox 1923	Consumer	Newark	Ohio	East	Late delivery	14/11/2021	O_ID_3000029	19/11/2021	Second Class	2	5	1.980000019	65.97000122	3	-3.650000095
C_ID_48434	Mary Selesnick	Office Supplies	Fellowes Stor/Drawer Steel Plus Storage Drawers	Corporate	Bossier City	Louisiana	South	Late delivery	15/08/2021	O_ID_3000030	16/12/2021	Second Class	2	5	16.30999947	95.97000122	3	38.22999954
C_ID_63936	Hannah Dahlen	Office Supplies	Tennsco 6- and 18-Compartment Lockers	Consumer	San Francisco	California	West	Shipping on time	25/07/2022	O_ID_3000031	27/07/2022	Second Class	2	2	30	299.9700012	3	132.2899933
C_ID_65030	Mary Selesnick	Technology	Plantronics S12 Corded Telephone Headset System	Corporate	Bossier City	Louisiana	South	Late delivery	11/07/2022	O_ID_3000032	16/08/2022	Second Class	2	6	74.98999786	299.9700012	2	76.04000092
C_ID_62571	Michael Sweed	Furniture	Sauder Cornerstone Collection Library	Corporate	Philadelphia	Pennsylvania	East	Shipping on time	10/02/2022	O_ID_3000033	10/04/2022	Second Class	2	2	12.60000038	179.9700012	2	43.52000046
C_ID_65922	Vincent Hackney	Technology	Logitech 910-002974 M325 Wireless Mouse for Web Scrolling	Consumer	Roswell	Georgia	South	Late delivery	23/08/2022	O_ID_3000034	26/08/2022	Second Class	2	3	18	179.9700012	2	40.49000168
C_ID_63936	Hannah Latchford	Furniture	DAX Black Cherry Wood-Tone Poster Frame	Consumer	Philadelphia	Pennsylvania	East	Shipping on time	25/07/2022	O_ID_3000035	27/07/2022	Second Class	2	2	10.5	150	2	20.93000031
C_ID_56448	Mary Bixby	Office Supplies	GBC Wire Binding Combs	Consumer	Seattle	Washington	West	Late delivery	17/04/2022	O_ID_3000157	23/04/2022	Standard Class	4	6	50	499.9500122	2	-269.9700012
C_ID_67892	Frances Latchford	Office Supplies	Round Specialty Laser Printer Labels	Consumer	Philadelphia	Pennsylvania	East	Late delivery	21/09/2022	O_ID_3000037	24/09/2022	Second Class	2	3	37.5	150	3	12.71000004
C_ID_56359	Megan Latchford	Technology	AT&T 841000 Phone	Consumer	Philadelphia	Pennsylvania	East	Late delivery	07/02/2022	O_ID_3000038	07/07/2022	Second Class	2	5	1.799999952	179.9700012	2	-285.0700073
C_ID_52477	Mary Miller	Office Supplies	Performers Binder/Pad Holder, Black	Corporate	Little Rock	Arkansas	South	Advance shipping	24/04/2022	O_ID_3000907	26/04/2022	Standard Class	4	2	5	499.9500122	5	55.93000031
C_ID_57567	Mary liebe	Office Supplies	Xerox 229	Home Office	Los Angeles	California	West	Advance shipping	22/04/2022	O_ID_3000909	24/04/2022	Standard Class	4	2	15	499.9500122	5	126.0899963
C_ID_53413	Grace Swindell	Office Supplies	GBC Standard Recycled Report Covers, Clear Plastic Sheets	Consumer	Scottsdale	Arizona	West	Late delivery	21/02/2022	O_ID_3000041	25/02/2022	Second Class	2	4	27	179.9700012	2	47.88000107
C_ID_56973	Mary Swindell	Office Supplies	Tyvek Top-Opening Peel & Seel Envelopes, Plain White	Consumer	Scottsdale	Arizona	West	Late delivery	07/11/2022	O_ID_3000042	18/04/2022	Second Class	2	4	6	150	3	23.46999931

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E-commerce Business Analysis

DELIVERY PERFORMANCE AND PROFITABILITY ANALYSIS

Key finding

On-time deliveries produce positive, stable profit; late deliveries are mixed and volatile; cancellations consistently generate loss.

Why it matters

Delivery failures aren’t only customer experience issues they are a major driver of profit leakage and should be prioritized for corrective action.

On-Time Deliveries
20.4K orders

Late Deliveries
62K orders

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KEY INSIGHTS AND PATTERNS IDENTIFIED

- **Volume ≠ profit:** High sales volume does not guarantee profitability when operational inefficiencies are present.
- **Top driver of losses:** Delivery cancellations were the single largest contributor to negative-profit orders.
- **Customer value stability:** Repeat customers provide more predictable value but remain sensitive to major service failures.
- **Concentrated risk:** Profitability issues were unevenly distributed, with certain regions and categories disproportionately affected—clarifying where management attention should focus to improve outcomes.

STRATEGIC RECOMMENDATIONS

- Prioritize improving delivery reliability for high-volume, low-margin categories, as these are most vulnerable to profit loss.
- Review and optimize shipping partners and methods in underperforming regions.
- Focus retention strategies on protecting repeat customers from service failures, given their importance to stable revenue.
- Reassess pricing and discount policies for thin-margin products to ensure they do not amplify losses when operational issues occur.

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EXPECTED BUSINESS IMPACT

Reduced loss from canceled orders	Better delivery performance lowers negative-profit transactions and protects margins.
Improved retention and trust	Fewer cancellations and clearer service reliability support repeat purchases over time.
Predictable profitability model	Visibility into operational drivers enables proactive management and sustainable business value.

REFLECTION AND FUTURE ANALYSIS OPPORTUNITIES

This case study demonstrates the value of business-oriented diagnostic analysis in an e-commerce context. With access to more granular logistics data, future analysis could evaluate shipping partner performance in greater detail and identify early indicators of delivery risk.

Additionally, integrating time-based trends could support predictive monitoring of profitability issues. This reflection highlights analytical maturity by acknowledging current limitations while identifying clear paths for deeper insight, mirroring how analytics work evolves in real organizational settings.

TOOLS & TECHNOLOGIES

A comprehensive overview of the software, platforms, and technologies I leverage to deliver data-driven insights and solutions.

Power BI

Google Sheets

tableau

SQL

X

TESTIMONIALS

Client from Khamsat

السلام عليكم ورحمة الله وبركاته، أتمنى أن تكوني بخير أستاذة نورهان. أتقدّم لكِ بجزيل الشكر والتقدير على تنفيذ العمل بشكل احترافي وسريع ودقيق، بل وأكثر مما هو متوقع. عملك يعكس جودة عالية واهتمامًا بالتفاصيل، وأسعدني جدًا التعاون معك في هذا المشروع. هذه ليست النهاية، بل بداية خير لأعمال قادمة بإذن الله. تقييمك من جانبي: ممتازة وأكثر واقيمك تقيم خاص حيث ان التقييم هنا من 5 فانتي تستحقين 10، مع خالص الشكر والتقدير

جودة الخدمة	★★★★★
التواصل والمتابعة	★★★★★
التسليم بالموعد	★★★★★

LET'S CONNECT & COLLABORATE

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THANK YOU!

I appreciate your time and interest in my portfolio. Let's connect and explore how data-driven insights can drive your success.

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